

Why We Must Automate Our Data Pipelines

A one-page brief for Operations | Prepared by Stanley Metone

The Risk: What Happens When a Person Types the Numbers In

Picture this: every night, someone on the team copies the day's sensor readings (temperature, pressure, and so on) from one spreadsheet into another, by hand, so the operations report is ready for the morning meeting. One night, they're tired and miss a row, or they paste a column one cell too low. Nobody notices, because the spreadsheet doesn't know what a "normal" reading looks like – it just accepts whatever is typed in. Three days later, a piece of equipment overheats because the report said everything was fine when it wasn't. That's not a hypothetical risk unique to us – it's the same failure mode behind well-known manual-entry disasters, like traders losing millions because someone fat-fingered a number into a spreadsheet with no checks in place. The problem is never that our people aren't careful. It's that manual copy-and-paste has no safety net.

The Solution: An Automatic Irrigation System, Not a Bucket Brigade

Think about watering a field. You could send workers out every day to carry buckets row by row – it works, but it's slow, and if someone skips a row, the plants there quietly die before anyone notices. Or you could install an automatic irrigation system: pipes that run on a schedule, sensors that shut the water off if pressure drops, and an alarm that rings if something's wrong – before the crop is damaged.

An automated data pipeline (a "pipeline" here just means a small program that moves and checks data on its own, on a schedule, without anyone typing it in) is the irrigation system for our numbers. Every night, it automatically pulls the day's readings, checks them against simple rules ("pressure can't be negative," "no reading can be missing an ID"), and only then loads the clean data into the report. If a rule is broken, it stops and flags it – instead of quietly passing bad numbers into the morning meeting.

The Value: What This Actually Buys Us

Time saved	~45 minutes of manual entry per day recovered → roughly 16 work-days a year, per person, given back
Error reduction	Every record is checked against the same rules, every time. The kind of one-off human slip that causes
Faster decisions	The morning report is ready the moment the pipeline finishes overnight – not whenever someone gets th

Bottom line: Excel isn't the problem – typing the same numbers into it by hand, every day, with no one checking the work, is. Automation doesn't replace the team's judgment; it makes sure the numbers they're making decisions on are actually correct.